Available online at www.sciencedirect.com
SCIENCE DIRECT®
Computer Coupling of Phase Diagrams and Thermochemistry 30 (2006) 171–178
www.elsevier.com/locate/calphad

Thermodynamic analysis of the stable and metastable Co–Cu and Co–Cu–Fe phase diagrams

Mauro Palumbo[a,*], Stefano Curiotto[a,b,c], Livio Battezzati[a]

[a] *Dipartimento di Chimica I.F.M., Centro di Eccellenza NIS, Università di Torino, via Giuria 7/9, 10125 Torino, Italy*
[b] *Materials Research Department, AFM 228, Risø National Laboratory, Frederiksborgvej 399, DK-4000 Roskilde, Denmark*
[c] *Niels Bohr Institute, Nanoscience Centre, University of Copenhagen, Copenhagen, Denmark*

Received 9 September 2005; received in revised form 5 October 2005; accepted 26 October 2005
Available online 16 November 2005

Abstract

The binary Co–Cu and the ternary Co–Cu–Fe systems have been optimized using the CALPHAD approach. Experimental data that have appeared recently in the literature have been used to improve the description of the metastable miscibility gap on both binary and ternary liquid phases. In the Co–Cu–Fe system, ternary parameters have been obtained to properly describe the miscibility gap. A significant improvement with respect to previous thermodynamic descriptions has been achieved. A set of model parameters for the liquid, f.c.c., and b.c.c. phases is given.
© 2005 Elsevier Ltd. All rights reserved.

Keywords: Co–Cu; Co–Cu–Fe; Metastable miscibility gap; Phase diagram

1. Introduction

Some alloys with a large positive enthalpy of mixing and nearly flat liquidus curves present a metastable miscibility gap in the undercooled liquid state. This behaviour has been studied extensively in both the Co–Cu and Cu–Fe systems and more recently in the ternary Co–Cu–Fe system, also under microgravity conditions, because of the fascinating prospect of producing materials containing finely dispersed phases for various applications [1].

Several thermodynamic assessments are available in the literature for both the binary Co–Cu and the ternary Co–Cu–Fe systems. The first reliable calculations of the binary Co–Cu phase diagram date back to the works by Kaufman [2] and Hasebe et al. [3]. Elder et al. [4] revised the phase diagram including experimental data on the composition of phases quenched from the liquid, as obtained with electron microprobe analysis. Gente et al. [5] reported the parameters derived in an unpublished assessment mainly based on the work of Hasebe et al. [3]. More recently, Kubista et al. [6] and Hari Kumar et al. [7] revised the thermodynamic description of the system. Thermodynamic optimizations of the ternary phase diagram have been published by Bamberger et al. [8] and by Wang et al. [9]. Cluster Variation Method (CVM) calculations have also been carried out by Bein et al. [10] and Antoni-Zdziobek et al. [11].

New accurate data on the metastable miscibility gap in the binary Co–Cu [12,13] and ternary Co–Cu–Fe [14] systems have been determined, showing that descriptions of the miscibility gap of previous calculations were no longer satisfactory. Thus, in this work a reassessment of both the Co–Cu and Co–Cu–Fe phase diagrams has been carried out including the newly available experimental data, and special care has been paid to the description of the miscibility gap in the liquid phase.

2. Evaluation of experimental data

2.1. Binary Co–Cu system

Early measurements of phase diagram points of the binary Co–Cu system were performed by Konstantinov [15], Sahmen [16] and Hashimoto [17] using different techniques. Liquidus data from Konstantinov show large scatter and are probably rather inaccurate. Later, liquidus/solidus points were determined by Nakagawa [18] (magnetic susceptibility),

* Corresponding author. Tel.: +39(0)116707567; fax: +39(0)116707855.
E-mail address: mauro.palumbo@unito.it (M. Palumbo).

0364-5916/$ - see front matter © 2005 Elsevier Ltd. All rights reserved.
doi:10.1016/j.calphad.2005.10.007

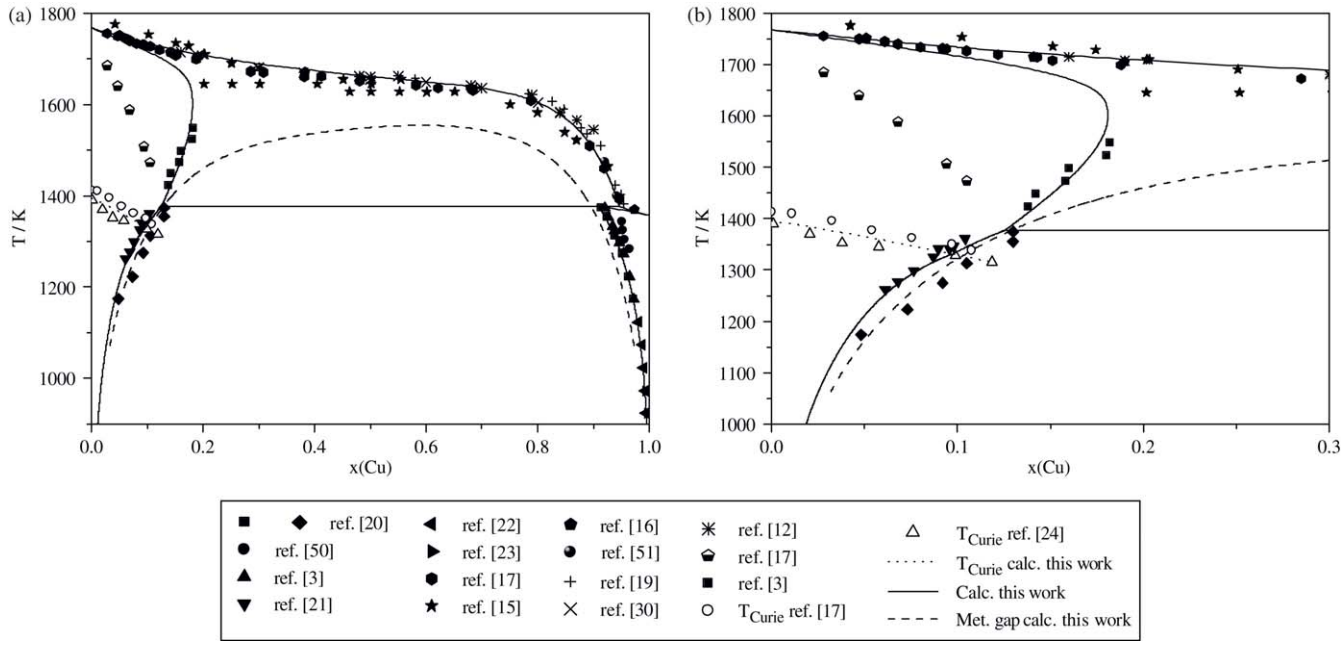

Fig. 1. (a) Calculated Co–Cu phase diagram (continuous line) and metastable miscibility gap in the liquid phase (dashed line) according to this work. Experimental data are also reported according to the literature sources listed in the inset. (b) Enlargement of the Co-rich part of the diagram.

Table 1
Thermodynamic and magnetic binary interaction parameters for the Co–Cu system evaluated in this study

Liquid phase	
G(LIQUID,CO,CU;0)	$+31\,500 - 1.3 * T$
G(LIQUID,CO,CU;1)	-595
G(LIQUID,CO,CU;2)	$+3700$
f.c.c. phase	
G(FCC_A1,CO,CU:VA;0)	$+34\,600 - 4.0 * T$
G(FCC_A1,CO,CU:VA;1)	$-6410 + 3.7 * T$
G(FCC_A1,CO,CU:VA;2)	$+4390$
TC(FCC_A1,CO,CU:VA;0)	$+812$
hcp phase	
G(HCP_A3,CO,CU:VA;0)	$+30\,000$

Lattice stabilities and magnetic properties of pure elements are taken from Ref. [38].

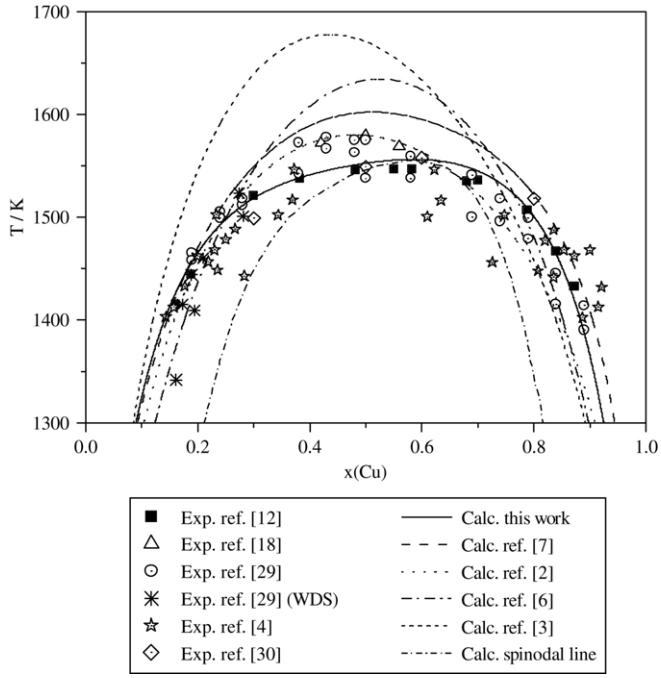

Fig. 2. Calculated Co–Cu metastable miscibility gap in the liquid phase according to various assessments (lines) and experimental points according to literature sources listed in the inset.

Hasebe et al. [3] (electron probe microanalysis) and Taskinen [19] (e.m.f. measurements). Two sets of solidus data in the Co-rich side of the diagram contradict each another, and this point will be further discussed in the following. The extent of solid solubility in the f.c.c. phase was determined by Hasebe et al. [3], Old et al. [20], Bruni et al. [21] (electron probe microanalysis), Livingston [22] (saturation magnetization) and Knappwost [23] (electrical resistivity). The Curie temperature of pure Co is decreased by the addition of Cu. The effect has been studied by Köster et al. [24] and Hashimoto [17].

Thermochemical data have also been determined by several authors. Dokken and Elliott [25], Iguchi et al. [26] and, more recently, Turchanin [27] measured the enthalpy of mixing of cobalt–copper alloys at 1473 K, 1823 K and 1834 K. Activity data were obtained by Timberg et al. [28] and Taskinen [19] using mass spectrometry and e.m.f. techniques, respectively.

The first investigation of the metastable miscibility gap was carried out by Nakagawa [18] by supercooling the liquid samples and measuring the change in magnetic susceptibility. A few points were obtained close to the maximum of the gap. Elder et al. [4] heated and cooled selected alloys in an electromagnetic levitation apparatus, monitoring the temperature with a pyrometer. Electron microprobe analysis was then used to determine phase compositions. Similar experiments were carried out by Robinson et al. [29]. Yamauchi et al. [30] studied the temperature change during melting and

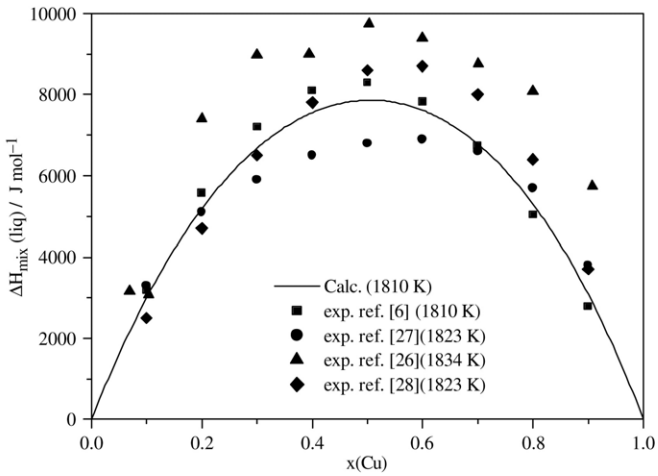

Fig. 3. Calculated enthalpy of mixing of the liquid phase in the Co–Cu system at 1810 K (continuous line) and experimental values (points). Reference states are pure elements in the liquid state at the same temperature.

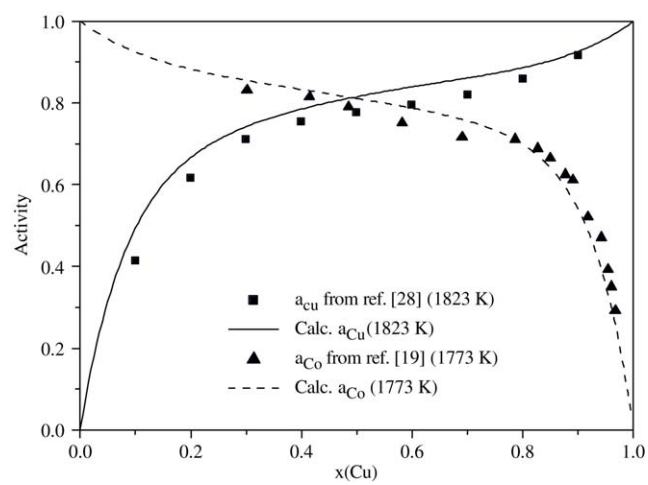

Fig. 4. Calculated activity of copper at 1823 K and of cobalt at 1773 K and experimental data in the Co–Cu system.

Table 2
Thermodynamic ternary interaction parameters for the Co–Cu–Fe system evaluated in this study

Liquid phase	
G(LIQUID,CO,CU,FE;0)	5760
G(LIQUID,CO,CU,FE;1)	−42 000
G(LIQUID,CO,CU,FE;2)	13 400
f.c.c. phase	
G(FCC_A1,CO,CU,FE:VA;0)	$-47\,900 + 20 * T$
G(FCC_A1,CO,CU,FE:VA;1)	$-47\,900 + 20 * T$
G(FCC_A1,CO,CU,FE:VA;2)	$-47\,900 + 20 * T$
b.c.c. phase	
G(BCC_A2,CO,CU,FE:VA;0)	$-114\,000 + 50 * T$
G(BCC_A2,CO,CU,FE:VA;1)	$-114\,000 + 50 * T$
G(BCC_A2,CO,CU,FE:VA;2)	$-114\,000 + 50 * T$

Lattice stabilities and magnetic properties of pure elements are taken from Ref. [38]. Parameters for the binary systems are taken from this work (Co–Cu, Table 1), Ohnuma et al. [42] (Co–Fe) and Jansson et al. [45] (Cu–Fe).

solidification of several Co–Cu alloys using a thermocouple directly inserted in the melt. Finally, Cao et al. [12,13,31] determined the binodal line over a wide composition range using high temperature calorimetry and a glass fluxing technique and thus obtained more accurate data on the liquid miscibility gap.

2.2. Ternary Co–Cu–Fe system

Several experimental data on the ternary system are available for phase diagram points, but no thermochemical measurements have been reported in the literature. Jellinghaus [32] and Maddocks et al. [33] first investigated vertical sections of the diagram using thermal analysis and metallography. Isothermal sections of the diagram were obtained by a number of authors [34,35,10,9,36] at several temperatures. Liquidus and miscibility gap points of several alloys were determined by Kim and Abbaschian [37] using a pyrometer to monitor the temperature and electromagnetic levitation to heat and cool the samples. Bamberger et al. [8] reported the liquidus temperatures of some alloys. Very recently, Cao et al. [14] applied high temperature DSC calorimetry to determine the liquidus points and the gap for a number of ternary compositions. In all these works the ternary miscibility gap was found to be metastable. It is noteworthy that the liquidus temperatures reported by Bamberger et al. are significantly different with respect to those reported by other authors. Some of these data are reported in the vertical sections in Fig. 7, showing differences with respect to the temperatures reported by other authors that are often more than 100 K. The other data sets are believed to be more reliable due to the experimental techniques employed, thus liquidus points from Bamberger et al. have been neglected in the optimization.

3. Thermodynamic modelling

3.1. Elements

The pure solid elements in their stable phases at 298.15 K were chosen as a reference state (SER, Standard Element Reference), unless otherwise specified. The Gibbs free energy of the pure elements (lattice stabilities) and the magnetic contributions have been taken from the PURE database [38].

3.2. Binary boundary systems

Thermodynamic modelling of binary and higher order phases has been performed according to the compound energy formalism [39]. The Gibbs free energy of a phase φ is thus given by the following equation:

$$G^{\varphi} = {}^{\mathrm{ref}}G^{\varphi} + {}^{\mathrm{id}}G^{\varphi} + {}^{\mathrm{ex}}G^{\varphi} + {}^{\mathrm{magn}}G^{\varphi} \quad (1)$$

where the first term refers to the weighted average of the Gibbs energy of pure elements, the second is the ideal mixing contribution, the third is an excess term that can be described using various models, and the last is the magnetic contribution to the Gibbs free energy.

The Co–Cu system presents three equilibrium phases: liquid, f.c.c. (γ) and hcp (ε). Both the liquid and f.c.c. phases have

a miscibility gap. Redlich–Kister polynomials have been used to describe these solution phases according to the following equations:

$$\begin{aligned} {}^{\mathrm{ref}}G^{\varphi} &= x_{\mathrm{Co}}G^{\varphi}_{\mathrm{Co}} + x_{\mathrm{Cu}}G^{\varphi}_{\mathrm{Cu}} \\ {}^{\mathrm{id}}G^{\varphi} &= RT[x_{\mathrm{Co}}\ln(x_{\mathrm{Co}}) + x_{\mathrm{Cu}}\ln(x_{\mathrm{Cu}})] \\ {}^{\mathrm{ex}}G^{\varphi} &= x_{\mathrm{Co}}x_{\mathrm{Cu}}L^{\varphi}_{\mathrm{Co,Cu}} = x_{\mathrm{Co}}x_{\mathrm{Cu}}\sum_{\nu=0}^{n}{}^{\nu}L^{\varphi}_{\mathrm{Co,Cu}}(x_{\mathrm{Co}} - x_{\mathrm{Cu}})^{\nu} \end{aligned} \tag{2}$$

where some of the ${}^{\nu}L^{\varphi}_{\mathrm{Co,Cu}}$ parameters are temperature dependent (Table 1).

No intermediate compound has been reported in this system.

The magnetic contribution has been expressed according to Inden [40] and Hillert and Jarl [41]:

$${}^{\mathrm{magn}}G^{\varphi} = RT\ln(\beta + 1)f(\tau) \tag{3}$$

where $f(\tau)$ is a polynomial expression containing T_c, the Curie temperature, and β is the mean atomic moment in Bohr magneton units. Both T_c and β can be functions of alloy composition as follows:

$$\begin{aligned} T_c &= {}^{0}T_c^{\mathrm{Co}}x_{\mathrm{Co}} + {}^{0}T_c^{\mathrm{Cu}}x_{\mathrm{Cu}} \\ &\quad + x_{\mathrm{Co}}\cdot x_{\mathrm{Cu}}\sum_{n}{}^{n}T_c^{\mathrm{CoCu}}(x_{\mathrm{Co}} - x_{\mathrm{Cu}})^{n} \\ \beta &= {}^{0}\beta^{\mathrm{Co}}x_{\mathrm{Co}} + {}^{0}\beta^{\mathrm{Cu}}x_{\mathrm{Cu}} \\ &\quad + x_{\mathrm{Co}}\cdot x_{\mathrm{Cu}}\sum_{n}{}^{n}\beta^{\mathrm{CoCu}}(x_{\mathrm{Co}} - x_{\mathrm{Cu}})^{n} \end{aligned} \tag{4}$$

where ${}^{0}T_c^{\mathrm{Co}}$, ${}^{0}T_c^{\mathrm{Cu}}$, ${}^{0}\beta^{\mathrm{Co}}$, ${}^{0}\beta^{\mathrm{Cu}}$ are the Curie temperatures and the mean atomic moments of pure elements (taken from the PURE database), whereas ${}^{n}T_c^{\mathrm{CoCu}}$ and ${}^{n}\beta^{\mathrm{CoCu}}$ are binary interaction parameters. Since no magnetic transition occurs in pure Cu, the relative values are nil. Furthermore, only one binary parameter has been considered for the Curie temperature in the present optimization (Table 1) owing to the limited amount of experimental data.

Binary models and parameters for the other binary systems have been taken from literature assessments. The Co–Fe phase diagram has been reviewed recently by Ohnuma et al. [42] and their parameters have been adopted in the present description of the ternary system. With respect to previous assessments, these authors improved the description of the low temperature part of the phase diagram by introducing new experimental data and a proper thermodynamic description of the chemical ordering in the b.c.c. phase. No miscibility gap is present in this system and the description of the high temperature equilibria is fully satisfactory.

Several assessments of the Cu–Fe system have been reported in the literature [43–46,3]. The existence of a metastable miscibility gap in the liquid phase has been recognized by various authors and measurements of the binodal line have been carried out [18,4,47,48]. A miscibility gap also occurs in the f.c.c. phase. The best agreement between the calculated and measured miscibility gap in the liquid is obtained using Jansson's parameters [45]. Although the assessment is older than others reported in the literature, recent experimental data have shown that it is reliable. Thus this set of values has been used in this work.

3.3. Ternary solution phases

Ternary liquid, f.c.c. and b.c.c. phases have been described using the conventional equation developed by Muggianu et al. [49]. Accordingly, the Gibbs free energy is expressed as follows:

$$\begin{aligned} {}^{\mathrm{ref}}G^{\varphi} &= x_{\mathrm{Co}}G^{\varphi}_{\mathrm{Co}} + x_{\mathrm{Cu}}G^{\varphi}_{\mathrm{Cu}} + x_{\mathrm{Fe}}G^{\varphi}_{\mathrm{Fe}} \\ {}^{\mathrm{id}}G^{\varphi} &= RT[x_{\mathrm{Co}}\ln(x_{\mathrm{Co}}) + x_{\mathrm{Cu}}\ln(x_{\mathrm{Cu}}) + x_{\mathrm{Fe}}\ln(x_{\mathrm{Fe}})] \\ {}^{\mathrm{ex}}G^{\varphi} &= x_{\mathrm{Co}}x_{\mathrm{Cu}}L^{\varphi}_{\mathrm{Co,Cu}} + x_{\mathrm{Co}}x_{\mathrm{Fe}}L^{\varphi}_{\mathrm{Co,Fe}} + x_{\mathrm{Cu}}x_{\mathrm{Fe}}L^{\varphi}_{\mathrm{Cu,Fe}} \\ &\quad + x_{\mathrm{Co}}x_{\mathrm{Cu}}x_{\mathrm{Fe}}(x_{\mathrm{Co}}\,{}^{0}L^{\varphi}_{\mathrm{Co,Cu,Fe}} \\ &\quad + x_{\mathrm{Cu}}\,{}^{1}L^{\varphi}_{\mathrm{Co,Cu,Fe}} + x_{\mathrm{Fe}}\,{}^{2}L^{\varphi}_{\mathrm{Co,Cu,Fe}}) \end{aligned} \tag{5}$$

where $L^{\varphi}_{\mathrm{Co,Cu}}$, $L^{\varphi}_{\mathrm{Co,Fe}}$, $L^{\varphi}_{\mathrm{Cu,Fe}}$ are binary interaction parameters as described in the previous section and ${}^{\upsilon}L^{\varphi}_{\mathrm{Co,Cu,Fe}}$ are ternary interaction parameters that can be temperature dependent.

No ternary compounds have been reported in the literature.

4. Results and discussion

All calculations have been performed using the ThermoCalc software package and the PARROT module for optimization.

4.1. Binary Co–Cu system

The binary Co–Cu phase diagram was first assessed, and the optimized parameters are reported in Table 1.

The calculated phase diagram is reported in Fig. 1 together with the corresponding experimental data reported in the literature. In general, good agreement has been obtained. As already pointed out, data from Konstantinov [15] are rather scattered and differ from other sets of data. Thus they are believed to be less accurate, and a low weight has been assigned to them in the optimization. Two conflicting data sets from Hashimoto [17] and Hasebe et al. [3] exist for the solidus line in the Co-rich side of the diagram. This fact has already been discussed by Kubista et al. [6], who annealed a $Co_{50.7}Cu_{49.3}$ sample at 1600 K for 1 h and then cooled it down to room temperature in order to establish which data set is more reliable. Their conclusion is that data from Hashimoto are correct. However, their calculated phase diagram is not fully in agreement with this data set. Furthermore, Hasebe et al. quenched their samples in ice brine, thus probably achieving a faster cooling rate. In the present optimization any attempt to obtain a better agreement with solidus data from Hashimoto led to a worsening of the agreement between calculated results and experimental data in other parts of the diagram. Thus, the data from Hasebe are thought to be more reliable, with the solubility being retrograde as for Cu–Fe.

In Fig. 2, the calculated miscibility gap is reported. Results according to other assessments in the literature and experimental data are also given for comparison. Data from Robinson et al. [29] and Elder et al. [4] are rather scattered.

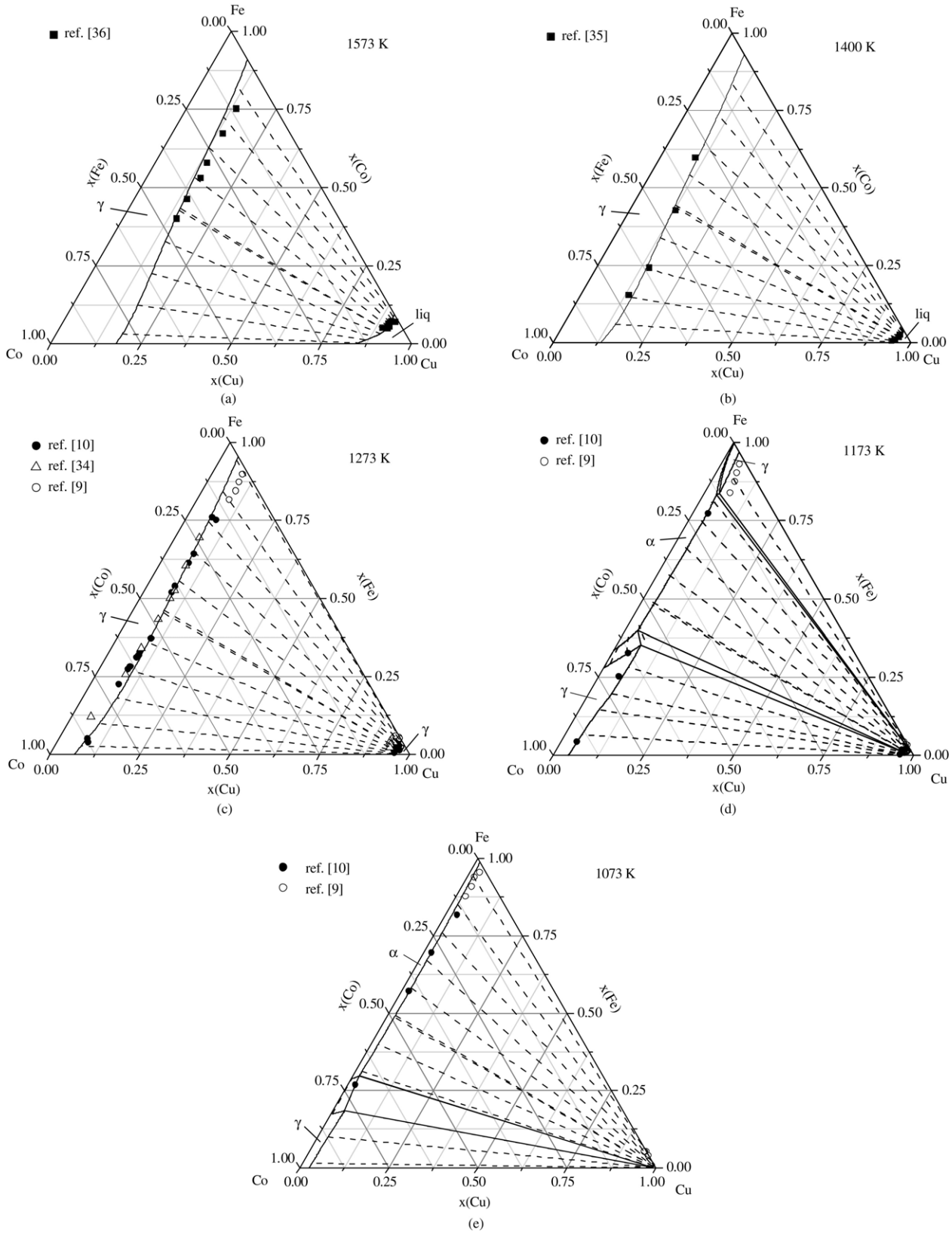

Fig. 5. Isothermal sections of the ternary Co–Cu–Fe system: (a) $T = 1573$ K; (b) $T = 1400$ K; (c) $T = 1273$ K; (d) $T = 1173$ K; and (e) $T = 1073$ K. Calculated phase boundaries: continuous lines; tie-lines: dashed lines; experimental data: points.

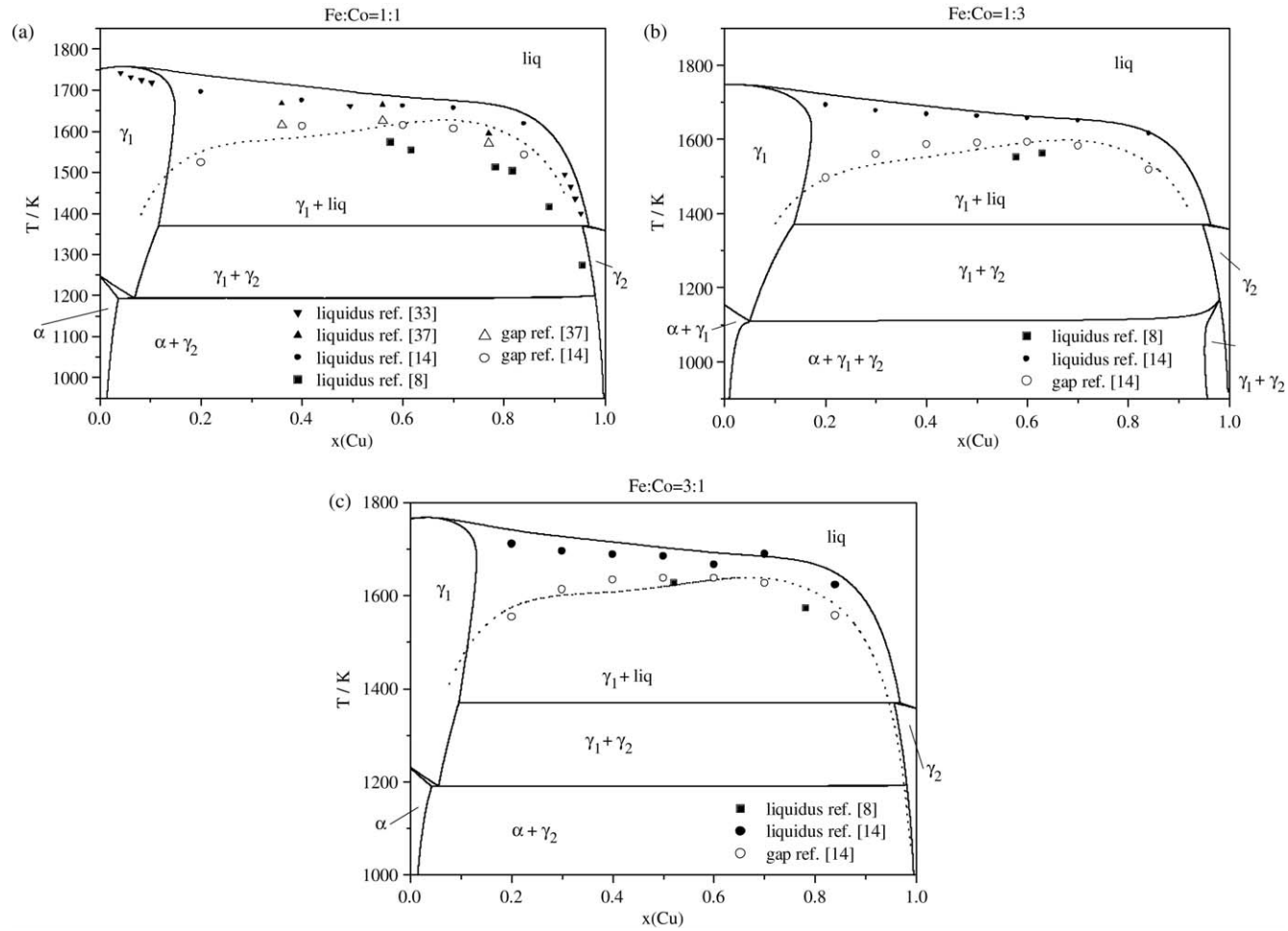

Fig. 6. Calculated vertical sections of the phase diagram (continuous line), including the metastable miscibility gap (dotted line), in the ternary Co–Cu–Fe system. Experimental points are also reported for comparison (black symbols refer to liquidus; open symbols refer to the metastable gap). (a) Fe:Co = 1:1 (at%); (b) Fe:Co = 1:3 (at%); (c) Fe:Co = 3:1 (at%).

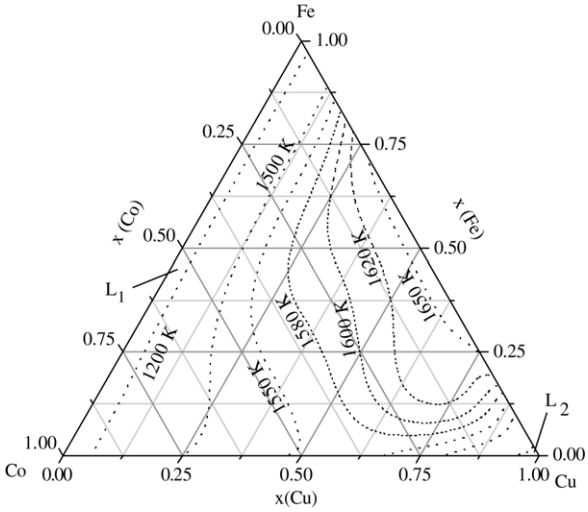

Fig. 7. Calculated metastable miscibility gap in the liquid phase at several temperatures in the Co–Cu–Fe system.

Their accuracy is implicitly limited by the experimental technique used by these authors with respect to data points from Cao et al. [12] obtained by high temperature calorimetry, which is believed to allow greater accuracy. It is evident that the description of the metastable miscibility gap has been improved significantly by our description. It is noteworthy that the maximum of the miscibility gap occurs at a lower temperature (1556 K and 58.5 at%) according to our new assessment in comparison with previous assessments. The minimum difference between the liquidus and the miscibility gap temperature occurs at 67.5 at% and is equal to 91 K.

A comparison with thermochemical data is reported in Figs. 3 and 4. The calculated enthalpy of mixing at 1810 K in Fig. 3 shows fairly good agreement with recent data from Kubista et al. [6] and Turchanin [27], whereas data from Iguchi et al. [26] present a larger difference. Activity data from Timberg et al. [28] are also converted and reported for comparison. Finally, in Fig. 4, calculated and experimental activities of Co and Cu are reported, with satisfactory agreement.

4.2. Ternary Co–Cu–Fe system

A first attempt to obtain a ternary description of the system by simply extrapolating binary parameters has been carried out. However, despite the new improved descriptions of the binary

systems now available with respect to the work of Bamberger et al. [8], we have obtained a stable miscibility gap in the liquid phase, as they did. As a consequence, ternary parameters have been added and optimized. Ternary parameters had to be considered for the f.c.c. and b.c.c. phases as well. Values of the assessed parameters for all phases are reported in Table 2. A ternary correction term was also introduced by Bamberger et al. [8] to eliminate the occurrence of the stable miscibility gap in the liquid phase. However, they estimated this term without performing an optimization including ternary data. Furthermore, they do not compare their results with data on the liquid–liquid separation temperatures.

On the contrary, Wang et al. [9] considered ternary parameters and experimental data on the liquid miscibility gap in their assessment, but they could not consider the new experimental data from Cao et al. [12,13], which appeared later. The present work also benefits from the new improved description of the binary Co–Cu system reported in the present work. Fig. 5 shows several sections of the ternary phase diagram calculated for different temperatures. The agreement between calculation and experiment is generally good, at both low and high temperatures. A slight deviation is obtained with respect to the data by Wang et al. [9] close to the Cu–Fe side of the diagram. This discrepancy can be related to the description of the solidus line in the Fe-rich side of the binary Cu–Fe phase diagram, which is not fully satisfactory. Furthermore, the CVM isothermal section at 1273 K calculated by Bein et al. [10] agrees very well with that reported in Fig. 5(c), whereas the CVM isothermal sections at 1173 K and 1073 K reproduce the correct shape of the phase diagram regions, but the Cu solubility in both α and γ phases is lower than in our calculations (Fig. 5(d), (e)). The biphasic α/γ region at 1073 K is slightly larger according to the CVM calculations.

Fig. 6 reports some vertical sections of the ternary phase diagrams corresponding to the experimental data by Cao et al. [14]. The agreement with the data relative to the metastable miscibility gap is good, whereas a slight overestimate of liquidus temperatures can be noticed, especially in the Fe:Co = 1:1 section. A similar systematic overestimation of liquidus temperatures was also obtained by Wang et al. Initially, we attributed the same weight to all data. Then, any attempt to improve the description of the liquidus line led to a worsening of the agreement between the calculated and experimental isothermal sections at high temperature. Data on the f.c.c. phase boundary in these isothermal sections are numerous from several authors and are in reciprocal agreement. Moreover, they cover a significant temperature range, while liquidus temperature data are limited to a narrow temperature range due to the flatness of the liquidus surface. It is therefore believed that this is the best possible description of the liquid phase to date.

The metastable miscibility gap appears to extend beyond the solidus in the Fe-rich part of the diagrams. There are limited data in this region and, therefore, the binodal line should be considered with caution.

Finally, Fig. 7 shows the calculated metastable miscibility gap in the liquid phase at several temperatures. It is noteworthy that the miscibility gap does not present a sharp saddle point in the middle of the diagram, as obtained in the assessment by Wang et al. [9]. This is mainly because of the new description of the binary Co–Cu system, where the maximum of the miscibility gap occurs at a lower temperature. This behaviour is clearly confirmed by the new data obtained by Cao et al. [14].

5. Conclusions

A new assessment of both the binary Co–Cu and ternary Co–Cu–Fe systems has been performed using the CALPHAD approach.

A significant improvement in the description of the miscibility gap in the liquid phase has been achieved in the binary Co–Cu system. In the Co–Cu–Fe system, ternary parameters had to be introduced in the description of the liquid phase in order to predict a metastable miscibility gap.

A fairly good agreement has been achieved between our thermodynamic description and experimental data in both the binary and ternary phase diagrams.

Acknowledgements

This work has been supported by the European Space Agency within the project “CoolCop” (ESA MAP Contract No. AO 99 - 010).

References

[1] I. Egry, D. Herlach, M. Kolbe, L. Ratke, S. Reutzel, C. Perrin, D. Chatain, Advanced Engineering Materials 5 (2003) 819.
[2] L. Kaufman, CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 2 (1978) 117.
[3] M. Hasebe, T. Nishizawa, CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 4 (1980) 83.
[4] S.P. Elder, A. Munitz, G.J. Abbaschian, Materials Science Forum 50 (1989) 137.
[5] C. Gente, M. Oehring, R. Bormann, Physical Review B 48 (1993) 13244.
[6] J. Kubista, J. Vrest'al, Journal of Phase Equilibria 21 (2000) 125.
[7] K.C. Hari Kumar, P. Wollants, Thermodynamic Optimization of the Co–Cu System, Presented at CALPHAD Conference, Grenoble, 1999.
[8] M. Bamberger, A. Munitz, L. Kaufman, R. Abbaschian, CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 26 (2002) 375.
[9] C.P. Wang, X.J. Liu, I. Ohnuma, R. Kainuma, K. Ishida, Journal of Phase Equilibria 23 (2002) 236.
[10] S. Bein, C. Colinet, M. Durand-Charre, Journal of Alloys and Compounds 313 (2000) 133.
[11] A. Antoni-Zdziobek, C. Colinet, Scandinavian Journal of Metallurgy 30 (2001) 265.
[12] C.D. Cao, G.P. Gorler, D.M. Herlach, B. Wei, Materials Science and Engineering A: Structural Materials Properties Microstructure and Processing 325 (2002) 503.
[13] C.D. Cao, T. Letzig, G.P. Gorler, D.M. Herlach, Journal of Alloys and Compounds 325 (2001) 113.
[14] C.D. Cao, G.P. Gorler, Chinese Physics Letters 22 (2005) 482.
[15] N. Konstantinov, Revue de Metallurgie 4 (1907) 983.
[16] R. Sahmen, Zeitschrift fur Anorganische Chemie 57 (1908) 1.
[17] U. Hashimoto, Journal of the Japanese Institute of Metals 1 (1937) 19.
[18] Y. Nakagawa, Acta Metallurgica 6 (1958) 704.
[19] P. Taskinen, Zeitschrift fur Metallkunde 73 (1982) 445.
[20] C.F. Old, C.W. Haworth, Journal of the Institute of Metals 94 (1966) 303.
[21] F.J. Bruni, J.W. Christian, Materials Science and Engineering 9 (1972) 241.

[22] J.D. Livingston, Transactions of the Metallurgical Society of AIME 215 (1959) 566.
[23] A. Knappwost, Z. Phys. Chem. 12 (1957) 30.
[24] W. Koster, E. Wagner, Zeitschrift fur Metallkunde 12 (1930) 230.
[25] R.N. Dokken, J.F. Elliott, Transactions of the Metallurgical Society of AIME 233 (1965) 1351.
[26] Y. Iguchi, Y. Tozaki, M. Kakizaki, S. Ban-ya, T. Fuwa, Journal of the Iron and Steel Institute of Japan 63 (1977) 953.
[27] M.A. Turchanin, Izvestiya Rossiiskoi Akademii Nauk. Metally 5 (1995) 12.
[28] L. Timberg, J.M. Toguri, T. Azakami, Metallurgical Transactions B 12B (1981) 275.
[29] M.B. Robinson, D. Li, T.J. Rathz, G. Williams, Journal of Materials Science 34 (1999) 3747.
[30] I. Yamauchi, N. Ueno, M. Shimaoka, I. Ohnaka, Journal of Materials Science 33 (1998) 371.
[31] C.D. Cao, D.M. Herlach, M. Kolbe, G.P. Gorler, B. Wei, Scripta Materialia 48 (2003) 5.
[32] W. Jellinghaus, Archiv fur das Eisenhuettenwesen 10 (1936) 115.
[33] W.R. Maddocks, G.E. Claussen, Iron and Steel Institute, London, Special Report 14, 1936, p. 116.
[34] M. Jiang, S.M. Hao, Proc. 6th Nat. Symp. Phase Diagram, Shenyang, 1990, p. 150.
[35] K. Oikawa, Thermodynamic Assessment of the Co–Cu-X (Cr, Mn, Fe, Ni) systems, Thesis of Master of Engineering, Tohoku University, Japan, 1981.
[36] H. Ohtani, H. Suda, K. Ishida, Isij International 37 (1997) 207.
[37] D.I. Kim, R. Abbaschian, Journal of Phase Equilibria 21 (2000) 25.
[38] SGTE Pure Elements database ver.3.00 updated from: Dinsdale AT CALPHAD, 1991, 5, pp. 317–325.
[39] M. Hillert, Journal of Alloys and Compounds 320 (2001) 161.
[40] G. Inden, Proc. Project Meeting CALPHAD V III, 1976, p. 41.
[41] M. Hillert, M. Jarl, CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 2 (1978) 227.
[42] I. Ohnuma, H. Enoki, O. Ikeda, R. Kainuma, H. Ohtani, B. Sundman, K. Ishida, Acta Materialia 50 (2002) 379.
[43] M.A. Turchanin, P.G. Agraval, I.V. Nikolaenko, Journal of Phase Equilibria 24 (2003) 307.
[44] Q. Chen, Z. Jin, Metallurgical and Materials Transactions A: Physical Metallurgy and Materials Science 26A (1995) 417.
[45] A. Jansson, Report D73 Metallografi KTH (1986) also in SSOL database, ThermoCalc AB.
[46] Y.Y. Chuang, R. Schmid, Y. Austin Chang, Metallurgical and Materials Transactions A: Physical Metallurgy and Materials Science 15A (1984) 1921.
[47] G. Wilde, R. Willnecker, R.N. Singh, F. Sommer, Zeitschrift fur Metallkunde 88 (1997) 804.
[48] G. Wilde, J.H. Perepezko, Acta Materialia 47 (1999) 3009.
[49] Y.M. Muggianu, M. Gambino, J.P. Bros, Journal de Chimie Physique 22 (1975) 83.